

Deletion of Tet proteins results in quantitative disparities during ESC differentiation partially attributable to alterations in gene expression.

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The Tet protein family (Tet1, Tet2, and Tet3) regulate DNA methylation through conversion of 5-methylcytosine to 5-hydroxymethylcytosine which can ultimately result in DNA demethylation and play a critical role during early mammalian development and pluripotency. While multiple groups have generated knockouts combining loss of different Tet proteins in murine embryonic stem cells (ESCs), differences in genetic background and approaches has made it difficult to directly compare results and discern the direct mechanism by which Tet proteins regulate the transcriptome. To address this concern, we utilized genomic editing in an isogenic pluripotent background which permitted a quantitative, flow-cytometry based measurement of pluripotency in combination with genome-wide assessment of gene expression and DNA methylation changes. Our ultimate goal was to generate a resource of large-scale datasets to permit hypothesis-generating experiments. We demonstrate a quantitative disparity in the differentiation ability among Tet protein deletions, with Tet2 single knockout exhibiting the most severe defect, while loss of Tet1 alone or combinations of Tet genes showed a quantitatively intermediate phenotype. Using a combination of transcriptomic and epigenomic approaches we demonstrate an increase in DNA hypermethylation and a divergence of transcriptional profiles in pluripotency among Tet deletions, with loss of Tet2 having the most profound effect in undifferentiated ESCs. We conclude that loss of Tet2 has the most dramatic effect both on the phenotype of ESCs and the transcriptome compared to other genotypes. While loss of Tet proteins increased DNA hypermethylation, especially in gene promoters, these changes in DNA methylation did not correlate with gene expression changes. Thus, while loss of different Tet proteins alters DNA methylation, this change does not appear to be directly responsible for transcriptome changes. Thus, loss of Tet proteins likely regulates the transcriptome epigenetically both through altering 5mC but also through additional mechanisms. Nonetheless, the transcriptome changes in pluripotent Tet2^{-/-} ESCs compared to wild-type implies that the disparities in differentiation can be partially attributed to baseline alterations in gene expression.

TET-mediated active DNA demethylation: mechanism, function and beyond.

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In mammals, DNA methylation in the form of 5-methylcytosine (5mC) can be actively reversed to unmodified cytosine (C) through TET dioxygenase-mediated oxidation of 5mC to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC), followed by replication-dependent dilution or thymine DNA glycosylase (TDG)-dependent base excision repair. In the past few years, biochemical and structural studies have revealed mechanistic insights into how TET and TDG mediate active DNA demethylation. Additionally, many regulatory mechanisms of this process have been identified. Technological advances in mapping and tracing the oxidized forms of 5mC allow further dissection of their functions. Furthermore, the biological functions of active DNA demethylation in various biological contexts have also been revealed. In this Review, we summarize the recent advances and highlight key unanswered questions.

Role of TET enzymes in DNA methylation, development, and cancer.

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The pattern of DNA methylation at cytosine bases in the genome is tightly linked to gene expression, and DNA methylation abnormalities are often observed in diseases. The ten eleven translocation (TET) enzymes oxidize 5-methylcytosines (5mCs) and promote locus-specific reversal of DNA methylation. TET genes, and especially TET2, are frequently mutated in various cancers, but how the TET proteins contribute to prevent the onset and maintenance of these malignancies is largely unknown. Here, we highlight recent advances in understanding the physiological function of the TET proteins and their role in regulating DNA methylation and transcription. In addition, we discuss some of the key outstanding questions in the field.

Mechanisms that regulate the activities of TET proteins.

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The ten-eleven translocation (TET) family of dioxygenases consists of three members, TET1, TET2, and TET3. All three TET enzymes have Fe²⁺ and α -ketoglutarate (α -KG)-dependent dioxygenase activities, catalyzing the 1st step of DNA demethylation by converting 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), and further oxidize 5hmC to 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC). Gene knockout studies demonstrated that all three TET proteins are involved in the regulation of fetal organ generation during embryonic development and normal tissue generation postnatally. TET proteins play such roles by regulating the expression of key differentiation and fate-determining genes via (1) enzymatic activity-dependent DNA methylation of the promoters and enhancers of target genes; and (2) enzymatic activity-independent regulation of histone modification. Interacting partner proteins and post-translational regulatory mechanisms regulate the activities of TET proteins. Mutations and dysregulation of TET proteins are involved in the pathogenesis of human diseases, specifically cancers. Here, we summarize the research on the interaction partners and post-translational modifications of TET proteins. We also discuss the molecular mechanisms by which these partner proteins and modifications regulate TET functioning and target gene expression. Such information will help in the design of medications useful for targeted therapy of TET-mutant-related diseases.

Tet proteins can convert 5-methylcytosine to 5-formylcytosine and 5-carboxylcytosine.

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5-methylcytosine (5mC) in DNA plays an important role in gene expression, genomic imprinting, and suppression of transposable elements. 5mC can be converted to 5-hydroxymethylcytosine (5hmC) by the Tet (ten eleven translocation) proteins. Here, we show that, in addition to 5hmC, the Tet proteins can generate 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC) from 5mC in an enzymatic activity-dependent manner. Furthermore, we reveal the presence of 5fC and 5caC in genomic DNA of mouse embryonic stem cells and mouse organs. The genomic content of 5hmC, 5fC, and 5caC can be increased or reduced through overexpression or depletion of Tet proteins. Thus, we identify two previously unknown cytosine derivatives in genomic DNA as the products of Tet proteins. Our study raises the possibility that DNA demethylation may occur through Tet-catalyzed oxidation followed by decarboxylation.

TET enzymes, DNA demethylation and pluripotency.

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Ten-eleven translocation (TET) methylcytosine dioxygenases (TET1, TET2, TET3) actively cause demethylation of 5-methylcytosine (5mC) and produce and safeguard hypomethylation at key regulatory regions across the genome. This 5mC erasure is particularly important in pluripotent embryonic stem cells (ESCs) as they need to maintain self-renewal capabilities while retaining the potential to generate different cell types with diverse 5mC patterns. In this review, we discuss the multiple roles of TET proteins in mouse ESCs, and other vertebrate model systems, with a particular focus on TET functions in pluripotency, differentiation, and developmental DNA methylome reprogramming. Furthermore, we elaborate on the recently described non-catalytic roles of TET proteins in diverse biological contexts. Overall, TET proteins are multifunctional regulators that through both their catalytic and non-catalytic roles carry out myriad functions linked to early developmental processes.

TET Enzymes in the Immune System: From DNA Demethylation to Immunotherapy, Inflammation, and Cancer.

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Ten-eleven translocation (TET) proteins are iron-dependent and α -ketoglutarate-dependent dioxygenases that sequentially oxidize the methyl group of 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC). All three epigenetic modifications are intermediates in DNA demethylation. TET proteins are recruited by transcription factors and by RNA polymerase II to modify 5mC at enhancers and gene bodies, thereby regulating gene expression during development, cell lineage specification, and cell activation. It is not yet clear, however, how the established biochemical activities of TET enzymes in oxidizing 5mC and mediating DNA demethylation relate to the known association of TET deficiency with inflammation, clonal hematopoiesis, and cancer. There are hints that the ability of TET deficiency to promote cell proliferation in a signal-dependent manner may be harnessed for cancer immunotherapy. In this review, we draw upon recent findings in cells of the immune system to illustrate established as well as emerging ideas of how TET proteins influence cellular function.

DNA repair and erasure of 5-methylcytosine in vertebrates.

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DNA methylation plays important roles in development and disease. Yet, only recently has the dynamic nature of this epigenetic mark via oxidation and DNA repair-mediated demethylation been recognized. A major conceptual challenge to the model that DNA methylation is reversible is the risk of genomic instability, which may come with widespread DNA repair activity. Here, we focus on recent advances in mechanisms of TET-TDG mediated demethylation and cellular strategies that avoid genomic instability. We highlight the recently discovered involvement of NEIL DNA glycosylases, which cooperate with TDG in oxidative demethylation to accelerate substrate turnover and promote the organized handover of

harmful repair intermediates to maintain genome stability.

The function and regulation of TET2 in innate immunity and inflammation.

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TET2, a member of ten-eleven translocation (TET) family as α -ketoglutarate- and Fe^{2+} -dependent dioxygenase catalyzing the iterative oxidation of 5-methylcytosine (5mC), has been widely recognized to be an important regulator for normal hematopoiesis especially myelopoiesis. Mutation and dysregulation of TET2 contribute to the development of multiple hematological malignancies. Recent studies reveal that TET2 also plays an important role in innate immune homeostasis by promoting DNA demethylation or independent of its enzymatic activity. Here, we focus on the functions of TET2 in the initiation and resolution of inflammation through epigenetic regulation and signaling network. In addition, we highlight regulation of TET2 at various molecular levels as well as the correlated inflammatory diseases, which will provide the insight to intervene in the pathological process caused by TET2 dysregulation.

DNA demethylation pattern of in-vitro fertilized and cloned porcine pronuclear stage embryos.

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Recent studies in mice showed that the Ten-eleven translocation Enzymes (TET) family is involved in the active DNA demethylation. The isotype TET-3 is responsible for the conversion of 5mc (5-methylcytosine) to 5hmc (5-hydroxymethylcytosine) at the pronuclear stages of mouse embryo. This study was performed to investigate the pattern of methylation change and the role of TET family in the demethylation process of porcine in-vitro fertilization (IVF) and somatic cell nuclear transfer (SCNT) derived embryo. Bisulfite-sequencing PCR (BSP) and DNA glucosylation and digestion before quantitative PCR (qGluMS-PCR) were done to evaluate the exact change of methylation during porcine pronuclear stages. The results showed that the amount of 5hmc detected increased whereas 5mc decreased in IVF embryo from pronuclear stage 2 (PN2) to pronuclear stage 5 (PN5). In addition, Immunofluorescent staining showed that the 5hmc signal, also detected in oocytes, significantly increased in both pronucleus from fertilization to PN2. The amount of 5hmc continued to rise in male pronucleus but decreased to a very low level in female pronucleus from PN2 to PN5. The above results indicate that female pronucleus might undergo active demethylation only at early pronuclear stages. On the other hand, male pronucleus might undergo active demethylation throughout all pronuclear stages. The expression of three TET isotypes (TET-1, TET-2, TET-3) were tested and TET-3 was found to be the highest expressed isotype. High TET-3 concentrations observed mainly in male pronucleus using immunofluorescent staining, implying that TET-3 might be the main enzyme which catalyzes the conversion of 5mc to 5hmc. In contrast, no TET-3 signal was detected in female pronucleus through the pronuclear stages. The demethylation pattern of SCNT embryos resembled that of the male pronucleus of IVF embryos, suggesting that active demethylation might happen in porcine cloned embryo.